

Tutorial 3 - Q3

Wednesday, September 24, 2025

2:07 PM

Question 3 Given the following MA processes:

a) $X_t = -0.9\epsilon_{t-1} + 0.2\epsilon_{t-2} + \epsilon_t,$

b) $X_t = 0.3\epsilon_{t-1} - 0.6\epsilon_{t-2} + \epsilon_t,$

c) $X_t = -1.5\epsilon_{t-1} + 0.75\epsilon_{t-2} - 0.125\epsilon_{t-3} + \epsilon_t,$

where $\{\epsilon_t\}$ is a WN. Show that all of these processes are invertible.

When is a general MA(q) invertible? See Thm 4.2.

Theorem 4.2 An MA(q) process X_t is invertible, if all of the roots of $\psi(z)$ lie outside the unit circle, i.e., iff

$$|z_i| > 1, \quad \forall 1 \leq i \leq q.$$

1. An MA(1) process $X_t = \psi\epsilon_{t-1} + \epsilon_t$ is invertible iff $|\psi| < 1$.
2. An MA(2) process $X_t = \psi_1\epsilon_{t-1} + \psi_2\epsilon_{t-2} + \epsilon_t$ is invertible iff all three of the following conditions hold:
 - (a) $\psi_1 + \psi_2 > -1$;
 - (b) $\psi_1 - \psi_2 < 1$;
 - (c) $-1 < \psi_2 < 1$.

(a) MA(2): $X_t = -0.9\epsilon_{t-1} + 0.2\epsilon_{t-2} + \epsilon_t$

(a) $\psi_1 + \psi_2 = -0.9 + 0.2 = -0.7 > -1$

(b) $\psi_1 - \psi_2 = -0.9 - 0.2 = -1.1 < 1$

(c) $-1 < (\psi_2 = 0.2) < 1$

$\therefore X_t$ is an invertible MA(2) process.

b

$$MA(2): X_t = 0.3 \epsilon_{t-1} - 0.6 \epsilon_{t-2} + \epsilon_t$$

$$(1) \quad \phi_1 + \phi_2 = 0.3 - 0.6 = -0.3 > -1$$

$$(2) \quad \phi_1 - \phi_2 = 0.3 + 0.6 = 0.9 < 1$$

$$(3) \quad -1 < (\phi_2 = -0.6) < 1$$

$\therefore X_t$ is an invertible MA(2) process

$$(c) \quad MA(3): X_t = -1.5 \epsilon_{t-1} + 0.75 \epsilon_{t-2} - 0.125 \epsilon_{t-3} + \epsilon_t$$

$$\psi(z) = \sum_{i=0}^3 \psi_i z^i \quad \leftarrow \text{Characteristic equation}$$

$$= \psi_0 z^0 + \psi_1 z^1 + \psi_2 z^2 + \psi_3 z^3$$

$$= 1 - 1.5z + 0.75z^2 - 0.125z^3$$

$$\text{Now set } \psi(z) = 0$$

$$1 - 1.5z + 0.75z^2 - 0.125z^3 = 0$$

$$(1 - 1/2 z)(1 - 1/2 z)(1 - 1/2 z) = 0$$

$$\therefore \text{ we have } z_1 = z_2 = z_3 = 2 > 1$$

$$\therefore X_t \text{ is an invertible MA(3) process.}$$